

GenAI Under Attack: Evaluating Enterprise Security Gaps Against Emerging AI Threat Vectors

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BACKGROUND

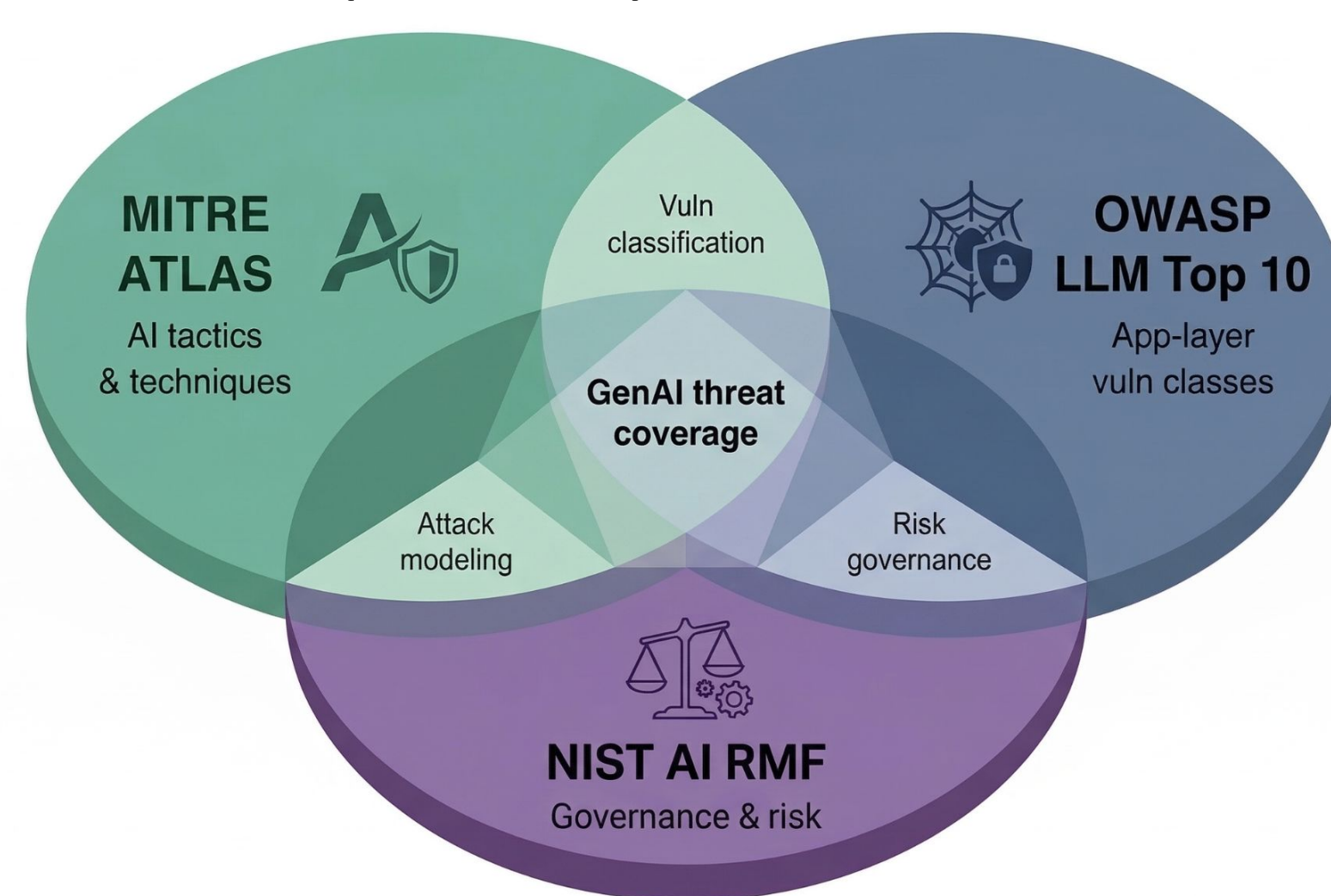
- A new technology stack introduces a new attack surface requiring additional defense methods (think: Cloud Computing, IoT, Containers).
- The integration of generative AI (GenAI) into production workloads has introduced new attack vectors that existing cybersecurity frameworks were not designed to adequately address.
- This project draws on GenAI vulnerability reports from 2022–2025 and early 2026, cross-referenced against OWASP LLM Top 10 and MITRE ATLAS classifications.
- This project evaluates which attack vectors current security controls leave most exposed, and which mitigation strategies most effectively close that gap.

OBJECTIVES

- Identify and categorize GenAI-specific attack vectors not adequately addressed by existing cybersecurity frameworks.
- Evaluate the coverage and gaps of three foundational frameworks of OWASP LLM Top 10, MITRE ATLAS, and NIST AI RMF, against the current GenAI threat landscape.
- Assess the effectiveness of existing security controls and tooling across key defensive dimensions including guardrails, model alignment, and access controls.
- Recommend a structured, layered mitigation strategy that organizations can apply across the GenAI deployment lifecycle.

FRAMEWORKS

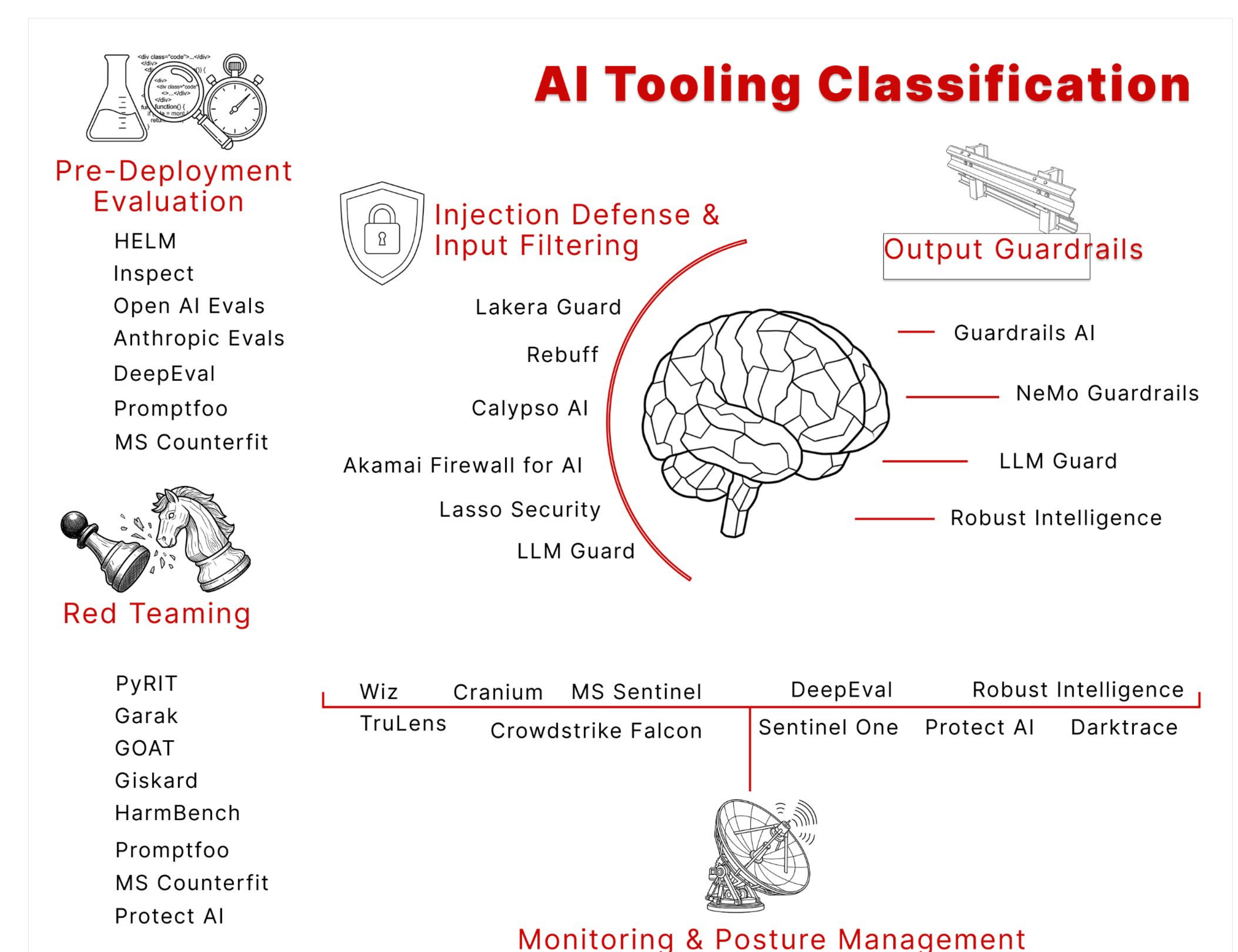
- MITRE Adversarial Threat Landscape for Artificial-Intelligence Systems (ATLAS).
- Open Worldwide Application Security Project (OWASP) Top 10 Web Application Security Risks.
- National Institute of Standards and Technology (NIST) AI Risk Management Framework (AI RMF).



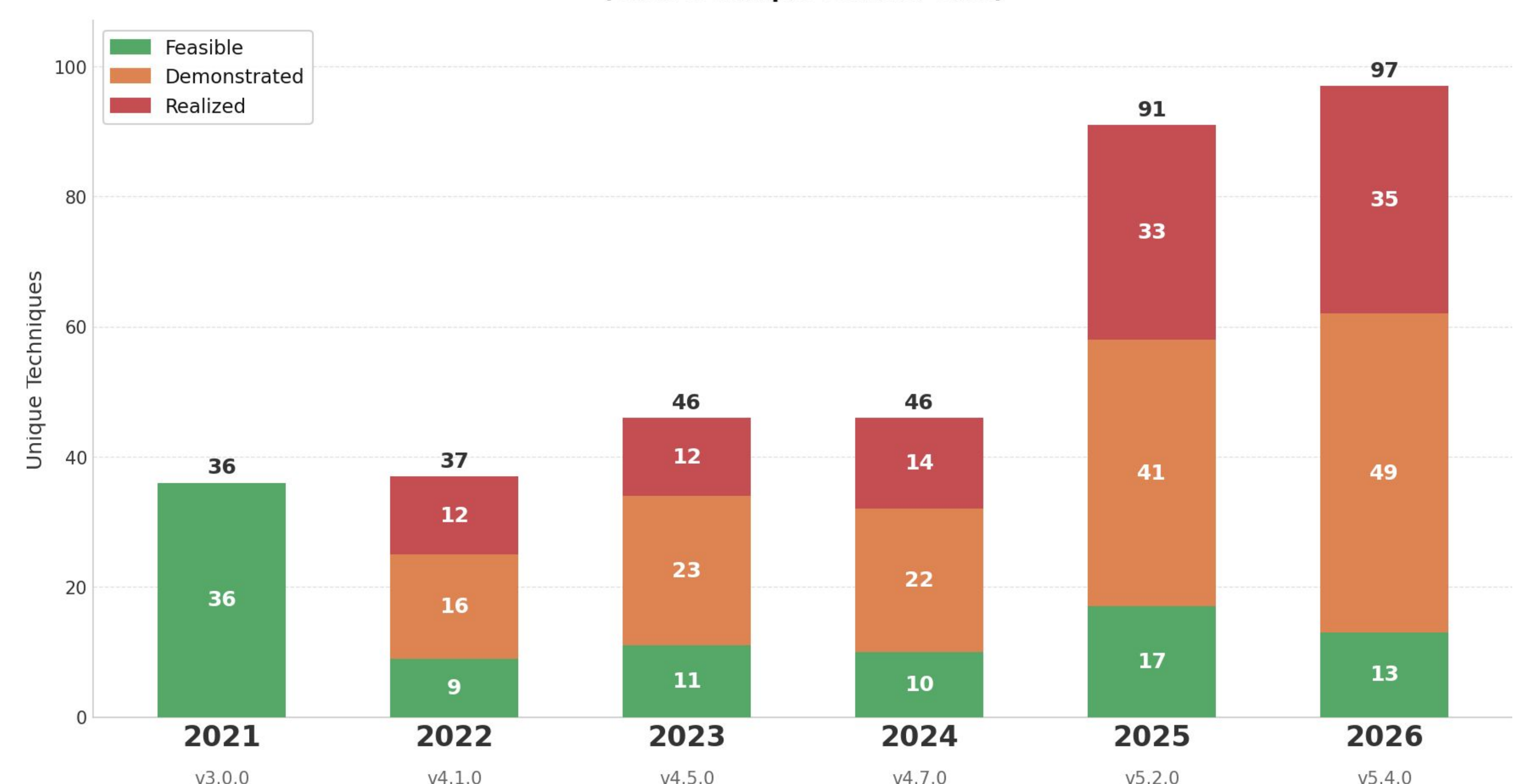
METHODOLOGY

- GenAI emerging threats and mitigation priorities were catalogued from vulnerability reports spanning 2022–2026
- Three foundational frameworks in GenAI security were evaluated.
- Security controls were assessed across six dimensions: guardrails, guidelines, model alignment, input/output validation, and least-privilege data access.
- Tools were evaluated across 29 publicly documented AI security products and grouped by defensive function and deployment stage.

RESULTS



ATLAS — All Techniques Maturity by Latest Release per Calendar Year (each technique counted once)



Rapid maturation from 36 techniques across 12 tactics in 2021 to 97 techniques across 16 tactics by March 2026

CONCLUSIONS

- Organizations require a structured response that combines framework-guided threat analysis, layered defensive tooling, and ongoing governance capable of adapting as emerging attack vectors.
- Pre-deployment red teaming emerged as the highest-leverage intervention, catching vulnerabilities that static analysis and runtime monitoring cannot detect.
- No single tool provides comprehensive coverage; evaluation revealed five functional categories: Pre-Deployment Evaluation, Red Teaming, Injection Defense & Input Filtering, Output Guardrails, and Monitoring & Posture Management.

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